

FAST BREEDER DOPPLER EFFECT INVESTIGATIONS IN ISRAEL*

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1. Introduction

Work on the Doppler effect in fast reactors was started in Israel in 1962. Since March 1964 this research has been partly supported by the KBB (Karlsruhe)–Euratom Association on Fast Reactors. Co-operation between the Karlsruhe and Israeli groups working in this field is very close, a few technical meetings a year ensuring scientific contact, comparison of results, change of emphasis, and co-ordination of effort.

Some aspects and results of this work have been summarized and reported recently.^(1–11)

The main effort was concentrated in four groups of problems:

- (a) computation methods;
- (b) temperature-dependent effective cross-sections;
- (c) parametric-type calculation of the Doppler effect on various systems;
- (d) special effects such as resonance interaction (fuel, coolant and structural material resonances), heterogeneity, multi-level formulae for fissile nuclides and others.

In this paper some results and some problems being investigated will be reviewed. New results, not reported previously, are also included.

2. Computation Methods

Three different methods of calculating the Doppler effect were developed and coded for the Philco-2000 computer. These are based on the successive- k , perturbation, and Monte Carlo techniques.

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Successive- k

Multi-group diffusion theory and the DIF 8 code, adapted from the Diffusion Code⁽¹²⁾ are used for the successive- k method. The system is brought to criticality at 300°K. When a certain convergence is obtained ($\bar{\epsilon} = 1 \times 10^{-6}$) and the system is critical ($k = 1$), changes in macroscopic cross-sections caused by a higher temperature are introduced and an iteration process on k is continued. When the prescribed convergence is obtained, the temperature is raised once more by introducing the necessary changes in effective cross-sections and the iteration process is continued. The calculation continues until the core reaches the highest temperature.

The choice of suitable input parameters and the reduction in loss of time involved in setup and managing of input as described in detail in ref. (5) made it possible to reduce the average running time for a criticality calculation to 12 sec, from a minimum running time of 4 min in previous calculations.

Perturbation

The perturbation code PERDIF⁽¹³⁾ developed originally for calculating such problems as material replacement experiments was adopted for the calculation of the Doppler effect of the same systems with the same nuclear input data (see below) so that a comparison of the two methods could be made. Comparative results are shown in Table 3.

Monte Carlo

While the successive- k and the perturbation methods have been used for calculating a relatively large series of systems, the Monte Carlo technique has only recently been used for calculating the Doppler effect. It is felt desirable, therefore, to give here a fuller description of the method adopted and the scope of the code.*

A Monte Carlo code, DOP has been written in FORTRAN for the Philco T-2000 computer to calculate for specific reactor configurations the average absorption and production in and the average leakage from the assembly in the energy interval $E_{\max}$ to $E_{\min}$ per neutron degraded past $E_{\max}$. In addition, the fraction of neutrons slowed down past $E_{\min}$ is also found. All these quantities are estimated at an initial temperature distribution in the reactor and their increments, calculated by a different Monte Carlo method, are determined when the temperatures are changed to any one of several subsequent distributions. An option has been included to evaluate in addition the derivatives with respect to temperature of the fractional absorption,

* It should be mentioned, however, that special effects such as resonance interaction and heterogeneity have been previously studied by us using a Monte Carlo approach as reported in refs. (2), (3) and (6).

production, leakage, and energy degradation past $E_{\min}$ at the initial temperature distribution.

If the assumption is made that the cross-sections are temperature independent at energies in excess $E_{\max}$ the Doppler change of the multiplication factor for a given alteration of the temperature distribution may be found by multiplying the change in production below $E_{\max}$ by the fraction of neutrons which are degraded past $E_{\max}$ per fission neutron produced in the assembly.⁽¹⁴⁾ The latter fraction must be obtained separately from a multi-group diffusion or S_N calculation.

The geometrical configurations dealt with by the code include concentric spheres and cylinders of finite height which may have radial and axial blankets and reflectors. In addition lattice configurations can be studied in which the core region consists of an infinite array of fuel pins of finite height. The fuel elements in the lattice, treated by the method given by Richtmeyer *et al.*⁽¹⁵⁾ may consist of a number of coaxial cylindrical regions. Blanket and reflector regions may be placed above and below the lattice and radial leakage allowed for by DB^2 terms.

The reactor constituents may include fissile and fertile materials in which the resonance parameters are selected from their statistical distributions, materials such as sodium with large scattering resonances, and structural materials. The measured resonance parameters of U^{238} are used at energies where they are available. Resonance cross-sections are calculated by the Breit Wigner single level formula using temperature dependent line shape functions. Contributions from resonance sequences with different spins J and parities, which can be obtained with the parts of the neutron wave corresponding to angular momenta $l = 0$ and $l = 1$, are included.

The neutrons are introduced into the assembly in accordance with specified spatial and energy distributions and their random walks followed from the instant they pass $E_{\max}$ until they reach $E_{\min}$ at the initial temperature distribution of the system. At other temperatures, corrections are applied to the neutron weights, in order to make the same path and collision points applicable to all temperature distributions.

The purpose of using only one set of random decisions at all temperature distributions in the reactor is to calculate differential effects directly for each path and at each collision and not as the final difference between two quantities each of which is subject to statistical errors possibly greater than the effect to be estimated. Nevertheless, other difficulties remain. The cross-section contributions increase with temperature in the wings of the resonances and decrease at their peaks while the asymmetric interference term between resonance and potential scattering, which decreases in the wings less slowly than the resonance cross-section itself, makes the temperature behaviour even more complicated.

Detailed tests are being made at present to determine whether the net

Doppler effect, i.e. the difference between its negative and positive contributions, can be determined with sufficient accuracy by the procedure outlined above or whether some procedure of sampling with expectation values should be introduced at each collision during the neutron histories. Some of the calculations together with some of the conclusions are reported in the supplement.

3. Nuclear Data

All calculations of the Doppler effect depend very heavily on nuclear parameters and basic resonance data. Lack of sufficient experimental information and the necessity to extrapolate low energy data to high energies with debatable or not fully justified methods, makes the problem of choosing a suitable nuclear physics input to the calculations a difficult problem of judgement. Also, there is a constant problem of re-evaluation to include the most recent experimental and theoretical information.

The uncertainties in the physics data are themselves difficult to assess, as is the influence of these uncertainties on the calculations.⁽¹⁶⁻¹⁹⁾

The nuclear parameters used in our calculations and the basis of their choice are given in ref. (6). A re-evaluation led to several changes as summarized in ref. (8) (TNSD-107, June 1965).

Presently preferred values together with the background for the changes are given in the Appendix. The methods of using the basic data for the Doppler calculations are described in refs. (6) and (4).

4. Systems Studied

The systems calculated to date include spherical 800–2500 litre reactors fuelled with plutonium and uranium in the metallic, oxide, and carbide forms. Different isotopic compositions of the plutonium were investigated. The above core volumes are in the range of the prototype fast reactors to be built in the United Kingdom (~ 250 MW(e)), the United States, Germany, and France. Also the Russian reactor BN-350 dual purpose power/desalination fast oxide breeder⁽²⁰⁾ being built near Shevchenko on the north-east coast of the Caspian Sea and having a core volume of 1870 litres is in the same core volume range. The core volumes studied are bigger, of course, than those of presently operating experimental fast reactors (U.K., U.S.A., U.S.S.R.) and smaller, on the other hand, than the contemplated 1000 MW(e) 5000–8000 litre cores.⁽²¹⁻²⁶⁾

An attempt has been made to find suitable one-dimensional models to two-dimensional reactor geometries for Doppler calculations. This was done to date for the 1000 MW(e) Karlsruhe-specified system.

The specification of the systems, together with the range of changes of some of the parameters is given in Table 1.

TABLE 1. SPECIFICATION OF SYSTEMS

1. Core volume	800, 1500 and 2500 litres
2. Core volume fractions	
Fuel and fertile material	0.25
Structural material	0.25
Coolant (Na) $S_0, S_1, \dots, S_5$	$= 0.50, 0.40, 0.30, 0.20, 0.10, 0$, respectively
3. Fuel composition	Density (g/cm ³)
Pu-U ²³⁸ (metal)	19
PuO ₂ -UO ₂ (oxide)	8.4
PuC-UC (carbide)	11.39
4. Coolant	Density (g/cm ³)
Liquid sodium	0.84
5. Plutonium	

Type	Atom per cent			
	Pu ²³⁹	Pu ²⁴⁰	Pu ²⁴¹	Pu ²⁴²
"A"	100	0	0	0
"B"	74.7	10.2	12.4	2.7
"C"	40	10	25	25

6. Blanket and reflector

	Thickness (cm)	Volume fractions		
		U ²³⁸	Na	Fe
Blanket	45	0.6	0.2	0.2
Reflector	30	0	0.4	0.6

7. Temperature $T_0, T_1, T_2, = 300, 900, 1500^\circ\text{K}$, respectively.

A fuller description of the systems together with a study of their static characteristics is given by Yiftah and Okrent.⁽²⁷⁾

5. Results

The calculation results using the successive- k method are summarized in Table 2.

Table 3 gives a comparison of the successive- k and perturbation methods for Pu "A" systems. The results are also depicted in Figs. 1, 2 and 3.

The main conclusions to be drawn from these results can be summarized

in terms of three effects, the volume, the fuel chemical composition, and the plutonium isotopic composition effects, as follows:

(1) The Doppler effect in the 800–2500 litre spherical cores described in this study is of the order of $10^{-5} \Delta k/^\circ\text{C}$ at 900°K and is sensitive to core volume. The sensitivity is largest in oxide fuel. The larger the core volume the greater is the absolute magnitude of the effect (see Fig. 1).

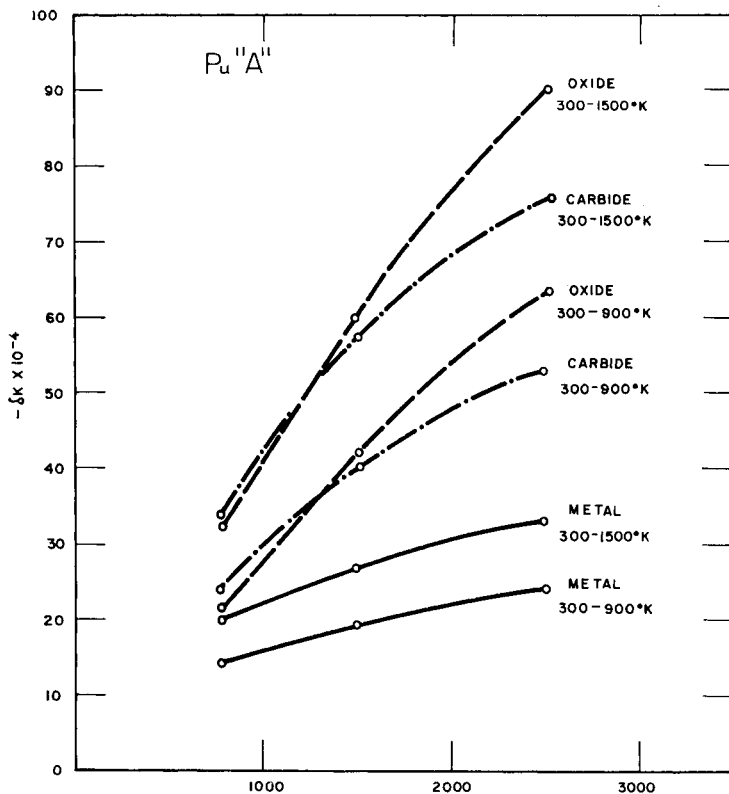


FIG. 1. Dependence of the Doppler effect on core volume for various chemical compositions of fuel (plutonium A).

(2) In reactors of equal core volumes the DE in volumes above 800 litres is smallest with metallic fuel and above 1400 litres is largest with oxide fuels (see Fig. 1).

(3) The presence of the high plutonium isotopes (Pu^{240} , Pu^{241} and Pu^{242}) in general increases the absolute value of the effect. Pu^{239} and Pu^{241} have a positive contribution while U^{238} , Pu^{240} and Pu^{242} contribute negatively. The overall effect depends on the relative contribution of every one of the isotopes present; it changes with Y^* , the ratio between the amounts of

$U^{238} + Pu^{240} + Pu^{242}$ and $Pu^{239} + Pu^{241}$ as seen in Table 2. It should be noted that the influence of the positive effect of Pu^{239} and Pu^{241} decreases with the increase of the content of Pu^{241} since the latter is a better fissile material than Pu^{239} (greater ν and smaller α) and thus the total mass of fissile isotopes for criticality is smaller if the plutonium contains greater amounts of high isotopes.⁽²²⁾ For a given core volume a decrease in the

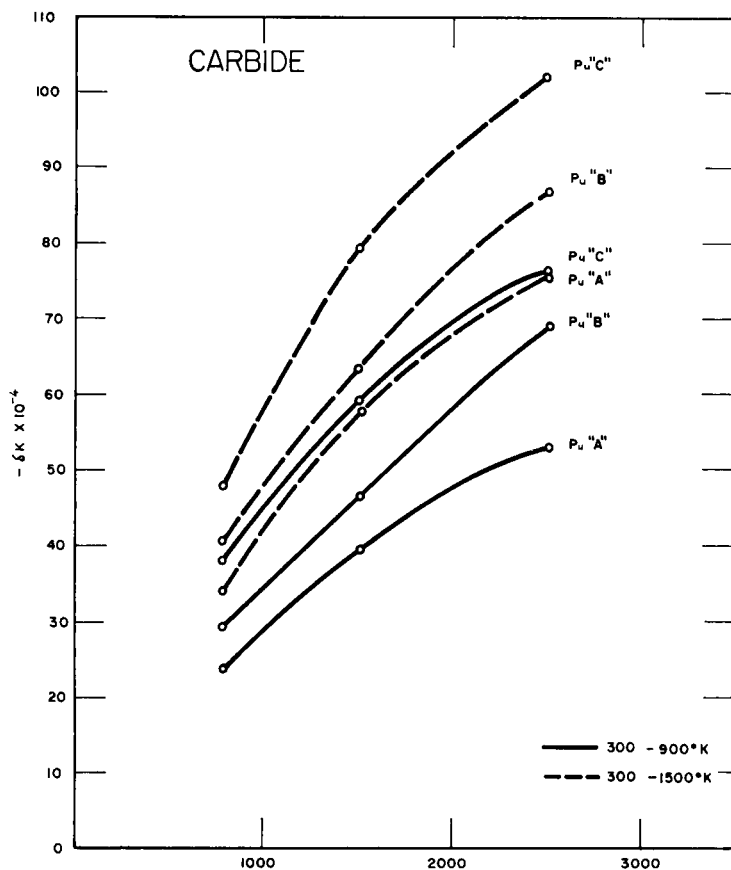


FIG. 2. Dependence of the Doppler effect on core volume and plutonium isotopic composition (chemical composition—carbide).

critical mass will imply an increase in the total mass of non-fissile material, contributing negatively to the overall effect. Altogether the effect of differences in the plutonium isotopic composition on the total Doppler coefficient may be as much as 30–40%. This is a larger effect than one could expect.

It should be mentioned that a Monte Carlo technique was used to study the effect of the interaction between the resonances of fertile and fissile

TABLE 2. RESULTS OF CALCULATIONS OF THE DOPPLER EFFECT, $\frac{\delta k}{k} \times 10^4$

Core volume (litres)	Fuel type	Pu "A"			Pu "B"			Pu "C"		
		Y*	300-900°K	300-1500°K	Y*	300-900°K	300-1500°K	Y*	300-900°K	300-1500°K
800	Metal Oxide Carbide	7.81	14.6	20.0	8.52	15.9	21.8	9.96	16.7	23.0
		2.98	21.8	32.5	3.34	24.5	34.8	4.04	37.6	50.0
		4.48	24.1	33.8	4.97	29.6	40.9	5.95	38.5	48.2
1500	Metal Oxide Carbide	9.38	19.7	26.7	10.22	21.2	29.3	11.90	22.4	30.9
		3.94	41.8	60.0	4.41	46.7	65.7	5.28	61.1	85.3
		5.63	39.7	57.4	6.24	46.8	63.8	7.41	59.3	79.0
2500	Metal Oxide Carbide	10.58	24.2	32.8	11.51	25.6	35.3	13.39	27.3	37.4
		4.76	63.3	89.8	5.29	81.2	110	6.32	86.3	129
		6.55	53.0	75.3	7.25	68.7	86.4	8.60	76.2	102

$$Y^* = \frac{U^{238} + Pu^{240} + Pu^{242}}{Pu^{239} + Pu^{241}}.$$

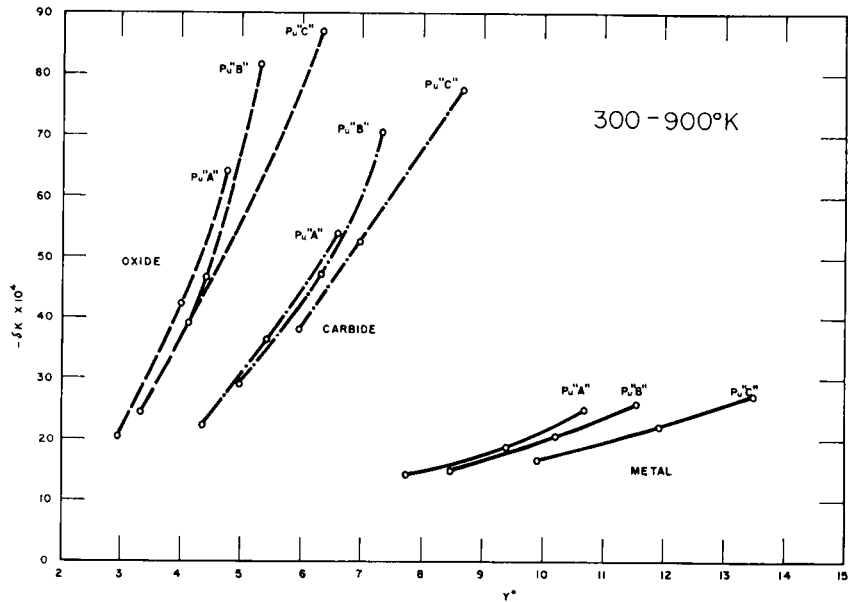


FIG. 3. Dependence of δk (300°-900°K) on isotopic composition and Y^* .

TABLE 3. COMPARISON OF THE SUCCESSIVE- k AND PERTURBATION METHODS OF CALCULATING THE DOPPLER EFFECT FOR Pu "A" SYSTEMS, $\frac{-\delta k}{k} \times 10^4$

Core volume (litres)	Core composition	300-900°K		300-1500°K	
		Perturbation method	Successive- k_{eff} method	Perturbation method	Successive- k_{eff} method
800	Metal	15.63	14.6	22.20	20.0
	Oxide	25.84	21.8	41.52	32.5
	Carbide	28.31	24.1	36.80	33.8
1500	Metal	21.31	19.7	30.30	26.7
	Oxide	50.96	41.8	79.3	60.0
	Carbide	47.85	39.7	75.72	57.4
2500	Metal	26.47	24.2	37.75	32.8
	Oxide	78.22	63.3	121.19	89.8
	Carbide	66.22	53.0	89.86* 101.37 75.599*	75.3

* Calculated using the perturbed flux in the perturbation code.

isotopes⁽³⁾ and the heterogeneity of fast reactor cores.⁽²⁾ The conclusions of these studies indicate that both these effects seem to be negligible.

The comparison of the successive-*k* and perturbation methods of computation (Table 3) shows that for large cores and for oxide and carbide fuels simple perturbation calculation is not adequate and one should use also perturbed fluxes in the calculation.

The problems under investigation are summarized in the supplement.

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6. Appendix 1

Nuclear Data for Doppler Calculations

Nuclear data for Doppler calculations are concerned with resonance parameters which are: the distance between levels, the neutron widths, the fission and capture widths. These are given as:

- (a) average values—which may depend upon the neutron energy, neutron angular momentum, and compound nucleus level spin;
- (b) statistical distributions about the averages—which may depend on the level spin but are not dependent on the neutron angular momentum and assumed non-dependent on energy in the range of the Doppler effect (i.e. <0.2 MeV).

Recent data which have direct bearing on average values are discussed, and certain values preferred (Table A1).

Recent views concerning the distributions are summarized (Table A2).

The strength function $\xi = \frac{\langle \Gamma_n^0 \rangle}{\langle D \rangle}$

The 1964 data of Garg *et al.*⁽¹⁾ of 230 U^{238} resonances yield a value $\xi_0 = (0.9 \pm 0.1) \times 10^{-4} \text{ eV}^{-\frac{1}{2}}$.

Assuming the same nuclear features for heavy nuclides having the same ground state spin, Garrison⁽²⁾ examined the question of two ξ_0 values for the two level sequences formed by $l = 0$ neutrons incident upon odd-even targets. His results may be summarized as follows: the two assumptions of a unique ξ_0 and D_J proportional to $(2J+1)^{-1}$ are together not in disagreement with experimental results.

For lack of experimental data it was commonly assumed that ξ_0 should be the same for all U and Pu isotopes—i.e. $\xi_0 \approx 10^{-4}$ —because of its dependence only on the nuclear radius. Today we have enough examples showing an appreciable effect upon the value of ξ_0 resulting from nuclear deformation. Such are: $\xi_0 = 0.72 \times 10^{-4}$ for Pu^{242} ,⁽²¹⁾ $\xi_0 = 1.54 \times 10^{-4}$ (in the energy range 100–200 eV) for $Pu^{239(18)}$ and $\xi_0 = 0.69 \times 10^{-4}$ for Th^{232} .⁽¹⁾

Also commonly assumed is equal values for the strength function of s and p wave neutrons. In support of this view is a value $\xi_s = 0.9 \times 10^{-4}$ for U^{238} ,

TABLE A1. PREFERRED AVERAGE NEUTRON RESONANCE PARAMETERS

	Fissile isotopes			Fertile isotopes		
	U ²³⁵	Pu ²³⁹	Pu ²⁴¹	U ²³⁸	Pu ²⁴⁰	Pu ²⁴²
$\langle D \rangle_0$, eV	0.7	2.5	1.3	19.5	11.8	18.1
$\langle D_{I-\frac{1}{2}} \rangle / \langle D_{I+\frac{1}{2}} \rangle$	1.0	3.4	1.4			
$\langle D(E) \rangle$ $\langle D_{th} \rangle$		0.98 0.90 0.81 0.66			0.98 0.88 0.80 0.65	
$\S_0$, eV ⁻¹	0.9×10^{-4}	1.15×10^{-4}	0.9×10^{-4}	0.9×10^{-4}	0.9×10^{-4}	0.72×10^{-4}
$\S_1$, eV ⁻¹	0.9×10^{-4}	2.5×10^{-4}	0.9×10^{-4}	0.9×10^{-4}	0.9×10^{-4}	0.9×10^{-4}
$\langle \Gamma_f \rangle \langle E \rangle$, mV		110 174 376 481 517 542	980 735 870 880 980			
$\langle \Gamma_f \rangle_{I-\frac{1}{2}} / \langle \Gamma_f \rangle_{I+\frac{1}{2}}$		6 (for low energy)	4 (for low energy)			
$\langle \Gamma_f \rangle$, mV	35	38	43	24	34	

given by Thomas and Bollinger,⁽¹⁹⁾ but in contradiction to it is the value $\xi_2 = 2.5 \times 10^{-4}$ recommended by Schmidt⁽¹⁷⁾ due to the findings of Uttley. ξ values are summarized in Table A1.

TABLE A2. PREFERRED STATISTICAL DISTRIBUTIONS

	Fissile isotopes	Fertile isotopes
$P(D(J))$	Wigner	Wigner
$P(\Gamma_f(J))$	$\chi^2_v; v = 1-3$	$\chi^2_v; v = 1-3$
$P(\Gamma_p^0)$	$\chi^2_v; v = 1$	$\chi^2_v; v = 1$
$P(\Gamma_\gamma)$	Very narrow	Very narrow

Definition of the symbols appearing in Tables A1 and A2

- | | |
|--|--|
| (1) $\langle D \rangle_0$ | The observed average level spacing of low energy resonances. |
| (2) $\langle D_{l-\frac{1}{2}} \rangle; \langle D_{l+\frac{1}{2}} \rangle$ | The average level spacing for the two equi-spin resonance sequences formed by $l=0$ neutrons incident on even-odd targets. |
| (3) $\frac{\langle D(E) \rangle}{\langle D_{th} \rangle}$ | The ratio between the average level spacing at energy E , and the corresponding spacing at low energies. |
| (4) Γ_f | The average fission width. |
| (5) Γ_γ | The average radiative capture width. |
| (6) Γ | The reduced neutron width. |
| (7) $P(x)$ | The statistical distribution of x . |
| (8) ξ_0 | The strength function for $l = 0$ neutrons. |
| (9) ξ_1 | The strength function for $l = 1$ neutrons. |

The Observed Level Spacing at Low Energies

The first step in the evaluation of $\langle D_J \rangle$ —the level spacing in a certain level sequence—is the determination of $\langle D \rangle_0$ —the observed level spacing. Since resonances are observable only at low energies, mostly $l = 0$ neutrons are concerned. In even-even nuclides, where $I = 0$, the observed series will be nearly identical to the $J = \frac{1}{2}$ sequence, whereas in even-odd nuclides, where $I \neq 0$, it will be a combination of the two $I + \frac{1}{2}$ and $I - \frac{1}{2}$ sequences.

Seventeen per cent of the 230 resonances of U^{238} , separated by Garg *et al.*,⁽¹⁾ are reported as uncertain. Not counting any of them yields $\langle D \rangle_0 = 20.8$ eV; counting them all gives $\langle D \rangle_0 = 17.7$ eV. Possibly the best choice would be between these two, i.e.

$$\langle D \rangle_0(\text{U}^{238}) = 19.5 \text{ eV.}$$

A separation of Pu^{239} resonances in the range 100–300 eV was recently published.⁽¹⁸⁾ A value $\langle D \rangle_0 = 2.5$ eV is found from this list of resonances, taking into account a remark by Schmidt⁽¹⁷⁾ that possibly 15% of the levels

are missing. This may be compared to a value of 2.8 eV resulting from the 19 low lying resonances separated by Bollinger *et al.*⁽⁴⁾

According to Moore *et al.*,⁽⁵⁾ twenty-one resonances of Pu²⁴¹ are certain in the range 4.3–3.9 eV, but three more very likely exist in this range. This results in

$$\langle D \rangle_0 (\text{Pu}^{241}) = 1.3 \text{ eV.}$$

For Pu²⁴⁰ nothing better can be suggested than the value adopted by Shaviv:⁽³⁾

$$\langle D \rangle_0 (\text{Pu}^{240}) = 11.8 \text{ eV.}$$

For U²³⁵ a result by Bowman *et al.*⁽⁶⁾ dated 1963 is $\langle D \rangle_0 = 0.64 \text{ eV}$. Based on a larger list of resonances, as given recently by Rainwater *et al.*,⁽²⁰⁾ is a value $\langle D \rangle_0 = 0.7 \text{ eV}$.

A value of $\langle D \rangle_0 = 18.1 \text{ eV}$ is derived by Pattenden⁽²¹⁾ from his separation of few Pu²⁴² resonances, recently reported.

$\langle D \rangle_0$ values are summarized in Table A1.

The Average Spacing in a Sequence of Equi-spin Levels

Levels with different spins belong to different statistical populations. The evaluation of $\langle D_{J_2} \rangle$ for a J_2 sequence, rests on the approximate law⁽²⁾

$$\frac{\langle D_{J_2} \rangle}{\langle D_{J_1} \rangle} = \frac{2J_1 + 1}{2J_2 + 1} \quad (1)$$

and a predetermined $\langle DJ_1 \rangle$ for $J_1 \neq J_2$.

In the case of low resonances in even–even targets we have, as discussed before,

$$\langle D_{\frac{1}{2}} \rangle = \langle D \rangle_0 \quad (\text{for even–even targets}).$$

In the case of odd–even targets (1) is used together with

$$\frac{1}{\langle D_{I+\frac{1}{2}} \rangle} + \frac{1}{\langle D_{I-\frac{1}{2}} \rangle} = \frac{1}{\langle D \rangle_0} \quad (2)$$

to evaluate $\langle D_{I+\frac{1}{2}} \rangle$ and $\langle D_{I-\frac{1}{2}} \rangle$.

In his extensive work⁽²⁾ Garrison compares (1) with experiment. To have good statistics, he does not distinguish between isotopes with the same ground-state spin and obtains the general result independent of the particular isotope under consideration, that (1) is consistent with experiment within the limits of experimental accuracy, which is not high.

In the present investigation an attempt was made to separate the two sequences of U²³⁵, Pu²³⁹, and Pu²⁴¹, the separation being based respectively

upon Bowman *et al.*,⁽⁶⁾ Fraser and Schwartz,⁽⁷⁾ and Moore *et al.*⁽⁵⁾ The resulting values for $\langle D_{I-\frac{1}{2}} \rangle / \langle D_{I+\frac{1}{2}} \rangle$ are compared with

$$\frac{I+1}{I} \left(= \frac{2J_1+1}{2J_2+1} \text{ for } l = 0 \text{ neutrons} \right) \text{ in Table A3.}$$

TABLE A3. $\langle D_{I-\frac{1}{2}} \rangle / \langle D_{I+\frac{1}{2}} \rangle$ FOR FISSILE ISOTOPES

Isotope	U ²³⁵	Pu ²³⁹	Pu ²⁴¹
Theoretical	1.3	3.0	1.4
Experimental	1.0	3.4	1.4

Variation of $\langle D \rangle$ with Energy

In the Fermi model of the nucleus⁽⁸⁾

$$\langle D(U) \rangle = U^n \exp(-2(aU)^{\frac{1}{2}})$$

where U is the excitation energy and the parameters n and a were originally given the value 2 and $A/11$ respectively.

Writing $U = E + B$, where E is the neutron energy and B the binding energy, and limiting ourselves to the Doppler range, i.e. $E \ll B$, it is found that

$$\frac{\langle D(E) \rangle}{\langle D_{\text{thermal}} \rangle} \simeq e^{-(a/B)^{1/2} E}$$

Recently two^(8,9) matches between experiment and theory were made in order to fix the value of a . Shaviv and Segev's⁽⁹⁾ value for a is based on less data than that of Buccino *et al.*,⁽⁸⁾ but has already proved itself in predicting inelastic distributions of neutrons.⁽⁹⁾

The values found for a for Pu and U isotopes are as follows:

Shaviv, Segev, and Yiftah: $a = 38$

Buccino *et al.*: $a = 12$

The fact that Buccino's value for a differs so much from that chosen by Shaviv is not surprising since a begins to rise sharply in the vicinity of $A = 240$. However, the data are so scarce that the point where the rise sets in cannot be determined very accurately. Recently Douglas and Barry⁽¹⁵⁾ also point to the fact that extremely different estimates for a are to be found in the literature. They recommend an intermediate value of $a = 30$.

Douglas and Barry: $a = 30$.

The preferred variation of $\langle D(E) \rangle$ is given in Table A1.

Average Observed Widths for Capture and Fission

Values of $\langle \Gamma_f \rangle$ are critically summarized by Garrison.⁽²⁾ They are quoted in Table A1.

Following YOM,⁽¹⁰⁾ Shaviv⁽³⁾ adopted the relation $\sigma_f(\text{Pu}^{241}) = \sigma_f(\text{U}^{233})$. Recent measurements by Smith *et al.*⁽¹¹⁾ and by Butler *et al.*⁽¹²⁾ above 20 keV of $\sigma_f(\text{Pu}^{241})$ led to the following comparison (Table A4) for $\langle \Gamma_f \rangle$ (Pu^{241}):

TABLE A4. $\langle \Gamma_f \rangle$ (IN mV) FOR Pu^{241}

Energy keV	Using $\sigma_f(\text{Pu}^{241}) = \sigma_f(\text{U}^{233})$	Using measured $\sigma_f(\text{Pu}^{241})$
20	817	980
50	606	735
100	810	870
150	889	880
200	1021	930

$\langle \Gamma_f \rangle (E)$ of Pu^{239} has been fitted with $\alpha(E)$ rather than with $\sigma_f(E)$.⁽³⁾ Some more data about $\alpha(E)$ are available now than in 1962 and the curve for $\alpha(E)$ recommended by Douglas and Barry⁽¹⁵⁾ differs somewhat from that recommended by Schmidt.⁽¹⁶⁾ $\langle \Gamma_f \rangle (E)$ by Shaviv⁽³⁾ (using Schmidt's recommendation) and by Douglas's recommendation follow in Table A5.

TABLE A5. $\langle \Gamma_f \rangle$ (IN mV) FOR Pu^{239}

Energy keV	Schmidt	Douglas
20	158	174
50	342	376
100	517	481

$\langle \Gamma_f \rangle$ values are summarized in Table A1.

Average Fission Widths of Equi-spin Level Sequences

As mentioned by Bowman *et al.*⁽⁶⁾ and by Garrison,⁽²⁾ the Bohr picture of the fission process leads to different $\langle \Gamma_f \rangle$ values for sequences with different spins, the lower $\langle \Gamma_f \rangle$ corresponding to the higher J .

The papers by Moore *et al.*⁽⁵⁾ and Bowman *et al.*,⁽⁶⁾ in which the combined observed resonance series is separated into its J components in Pu^{241} and

U^{235} , display a clear cut between two groups. In the one the Γ_f values are distributed about a lower $\langle \Gamma_f \rangle$ than in the other.

It was found that

$$U^{235}: \frac{\langle \Gamma_f \rangle (J=3)}{\langle \Gamma_f \rangle (J=4)} = 4.3$$

$$Pu^{241}: \frac{\langle \Gamma_f \rangle (J=2)}{\langle \Gamma_f \rangle (J=3)} = 4.3$$

Pu²³⁹

Uttley⁽¹⁸⁾ lists many resonances in the range 100–300 eV. Although Γ_f is not given in this list, it may be estimated in a way described in TNSD-114.⁽²²⁾ It is found that a greater part of the resonances are concentrated about $\Gamma_f = 40$ –50 mV than are scattered at higher Γ_f values. Assuming that the $J = 1$ sequence should be more populated than the $J = 0$ and that the superposition of two statistical populations should show some concentration around the sequence averages, the above aggregation may be identified as mainly $J = 1$ resonances. The $J = 0$ sequence must then have the average $\langle \Gamma_f \rangle = 300$ mV in order to get an overall weighted average of $\langle \Gamma_f \rangle = 110$ mV.

The Level Spacing Distribution

Let $x = \frac{D_J}{\langle D_J \rangle}$, where J denotes some level sequence of spin J ; let $P(x)$ be

the statistical distribution of x .

As extensively discussed by Garg,⁽¹⁴⁾ theoretical calculations of a refined nature and high resolution resonance data in even–even nuclei single out Wigner's formula

$$P_w(x) = \frac{\pi_x}{2} e^{-\frac{\pi_x^2}{4}}$$

as the only simple distribution which fits the experimental data well and is at the same time an excellent approximation to the more complicated theoretical distribution.

In odd–even targets the superposition of two uncorrelated sequences should give rise to higher probabilities for small spacings. However, such a superposition of two $P_w(x)$ overestimates the probability for small spacings when it is compared with experiment.⁽¹⁴⁾

One of two procedures may now be adopted—either a random distribution of two component resonance sequences narrower than $P_w(x)$ (e.g. the χ_μ^2 distribution with $\nu > 12$ as used in calculation of the Doppler effect)—or a correlated combination of two Wigner distributions which are based on the results of even–even nuclei and the refined theoretical predictions.

It is interesting to use the spin assignments in $\text{Pu}^{239(4,7)}$ and the proposed separations into component sequences in $\text{Pu}^{241(5)}$ and $\text{U}^{235(6)}$ to derive $P(x)$. Since in these cases the statistics are rather poor, we may assume $P(x)$ to be a χ^2 distribution leaving ν to be fixed by fitting with the data. The results are given in Table A6.

TABLE A6. VALUES OF ν

Sequence	$\text{U}^{235}J = 3$	$\text{U}^{235}J = 4$	$\text{Pu}^{239}J = 0$	$\text{Pu}^{239}J = 1$	$\text{Pu}^{241}J = 2$	$\text{Pu}^{241}J = 3$
ν	4	5	6	6-11	8-10	10

The average value $\langle \nu \rangle = 7$ —a value which makes the χ^2 distribution fit with $P_w(x)$.

Fission, Neutron Capture Width Distributions

The distribution of Γ_f in fissile elements is still a problem. Interference between the two sequences precludes a direct derivation of $P(\Gamma_f)$. It was therefore assumed to be a χ^2 distribution with ν degrees of freedom where ν is the number of open fission channels. However, this assumption implies that the number of open channels is the same for both spin states and that for a single spin state all channels are open to the same degree. These assumptions are, however, not well founded and should be eliminated. Consequently the value of ν is ambiguous. Furthermore, its determination by multi-level analysis is not always unique. It is accepted, however, that ν is limited to the range:

$$1 \leq \nu \leq 3 \text{ (for } P(\Gamma_f)\text{)}.$$

Recent analyses ^(1,2) confirm the view that if $P\langle \Gamma_n^0 \rangle$ is described by a χ_v^2 distribution the value

$$\nu \simeq 1 \text{ (for } P\langle \Gamma_n^0 \rangle\text{)}$$

fits the data best.

The distribution of Γ_γ is very narrow so that it would correspond to a very high value of ν in a χ_v^2 distribution.

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Supplement

Monte Carlo Calculations

Tests made with the Monte Carlo code to date have revealed some statistical difficulties in the estimation of the Doppler effect. The change in the multiplication factor of the assembly for a given temperature change is made up of positive and negative contributions which arise from the fact that the line shape functions decrease at the resonance peaks and increase in their wings when the temperature is raised. When the magnitudes of the contributions of opposite signs are not too large compared with their difference the net Doppler effect may be obtained with reasonable precision. An example of the statistics which has been obtained in the case of an assembly with an 800/ oxide core is given below. The system contained PuO_2 and UO_2 fuel with $\text{U}^{238}:\text{Pu}^{239}$ atomic ratio 3:1 and volume ratios of fuel structural material and sodium 1:1:2. In the blanket, 45 cm thick, U^{238} metal, coolant and structure were in the ratio 3:1:1. The reflector was made up of 3:2 parts structure and coolant by volume and was 30 cm thick. The results reported refer to histories run over a narrow energy interval from $E_{\text{max}} = 1.5$ keV to $E_{\text{min}} = 0.5$ keV, and give the fraction of neutrons produced in this energy range, W_p , at 300°K, and the change ΔW_p for a 100°K temperature increase, on which the Doppler effect depends (see Table S1 on p. 420).

TABLE SI. CALCULATION OF W_p AND DW_p (800 / OXIDE CORE)

Number of histories	W_p	ΔW_p	Number of histories	W_p	ΔW_p
10	0.260	0.0049	100	0.178	-0.0040
20	0.190	-0.0062	200	0.175	-0.0086
30	0.214	-0.0085	300	0.178	-0.0075
40	0.240	-0.0123	400	0.174	-0.0059
50	0.228	-0.0057	500	0.173	-0.0071
60	0.214	-0.0015	600	0.162	-0.0053
70	0.184	-0.0013	700	0.161	-0.0049
80	0.191	-0.0086	800	0.159	-0.0045
90	0.170	-0.0076	900	0.154	-0.0043
100	0.178 $\pm$ 0.030	-0.0040 $\pm$ 0.0053	1000	0.150 $\pm$ 0.009	-0.0036 $\pm$ 0.0013

It is clear that W_p can be estimated with reasonable precision in a not excessive number of histories. (The machine time for 1000 histories was about 10 min.) On the other hand the accuracy of ΔW_p is much poorer, the probable error giving the range in which the result is likely to lie with 50% certainty. The quantity ΔW_p is, in the above example, made up of negative contributions which exceed the positive contributions by about 30% in magnitude. In addition it should be noted that the results were calculated from cross-sections which included the effects of only one resonance above and one resonance below the neutron energy for each resonance sequence of Pu and U which can be excited by $l = 0$ neutrons. If contributions of the wings of more distant resonance were also included the positive and negative contributions became still nearer equal in magnitude with corresponding adverse effect on the statistics; the magnitude of ΔW_p after 1000 histories was still very much smaller than its probable error.

Similar results were obtained also for a large reactor assembly with a 6000l oxide core (U:Pu = 6.5:1) of composition 20%, 45%, and 35% fuel sodium and structure by volume and a blanket consisting of UO_2 sodium and structure in the ratio 2:2:1 by volume. In this case the first 100 histories yielded the following results:

TABLE S2. W_p AND DW_p FOR 600 l OXIDE CORE

Number of histories	W_p	ΔW_p for $\Delta T = 100^\circ K$	ΔW_p for $\Delta T = 1000^\circ K$
10	0.227	+0.0020	-0.0526
20	0.320	-0.0135	-0.1142
30	0.253	-0.0095	-0.0824
40	0.277	-0.0165	-0.0981
50	0.249	-0.0151	-0.0889
60	0.258	-0.0170	-0.0975
70	0.229	-0.0142	-0.0846
80	0.215	-0.0124	-0.0651
90	0.219	-0.0115	-0.0658
100	0.231 ± 0.031	-0.0078 ± 0.0036	-0.0573 ± 0.0161

Again the addition of the effect of more distant $l = 0$ resonances and also of $l = 1$ resonances made the statistics of ΔW_p still much poorer so that no conclusions appear to be possible of its magnitude (or even sign) and consequently of the Doppler effect in a reasonable amount of machine time. This conclusion becomes the more valid if one remembers that the accuracy is doubled at the expense of four times the calculation time, while in addition the range $E_{\max}$ to $E_{\min}$ over which calculations are made should be considerably widened.

It may also be pointed out that the capture fraction and its change with

temperature is predicted more accurately than the production fraction since it is much larger on account of the fact that it includes captures in fertile and other nuclei as well as in fissile nuclei.

Work is in progress at present with the aim of improving the statistics of the events in Pu^{239} by variance reducing techniques and consequently of the Doppler effect as far as possible.

Revision of Some Resonance Parameters

Recent data and resonance parameters evaluations⁽¹⁾ yield values 2×10^{-4} – 2.5×10^{-4} for ξ_1 of some fissile and fertile isotopes. It is felt, therefore, that such values should be adopted for all fissile and fertile isotopes. Accordingly, in Table A1, the following values should replace the corresponding ones:

	U^{235}	Pu^{239}	Pu^{241}	U^{238}	Pu^{240}	Pu^{242}
$\xi_1, \text{eV}^{-\frac{1}{2}}$	2.0×10^{-4}	2.5×10^{-4}	2.0×10^{-4}	2.5×10^{-4}	2.0×10^{-4}	2.0×10^{-4}

By determining the spins of 15 Pu^{239} resonances, Sauter and Bowman⁽²⁾ present an experimental basis for the theoretical assertion that small average fission widths are due to higher J values and vice versa. However, this relation concerns only the averages, and a clear-cut “boundary” between $J = 0$ fission widths and $J = 1$ ones⁽¹⁾ may only serve as a crude guess of $\langle \Gamma_f \rangle_J$. A boundary at 300 meV led Schmidt⁽¹⁾ to a value of 1.3 eV for $\langle \Gamma_f \rangle_{J=0}$; however, taking into account J assignments by Sauter and Bowman,⁽²⁾ a boundary at 150 meV may as well be tried. This leads to $\langle \Gamma_f \rangle_{J=0} = 1.0$ eV. Schmidt⁽¹⁾ regards the value of 1.3 eV as approaching his recommended value of 2.8 eV given by channel theory. Since, however, the question of the number of open channels is not at all settled, as, for instance, is clear from Garrison’s analysis,⁽³⁾ it is felt that a value in between 1.0 and 1.3 eV should be adopted. Accordingly in Table A1 the following revision is made:

$\langle \Gamma_f \rangle_{I-\frac{1}{2}} / \langle \Gamma_f \rangle_{I+\frac{1}{2}}$

Pu^{239}
17–23
(for resolved range)

Main Specific Problems under Investigation

The specific problems under investigation now include the following:

- (1) Development of a Monte Carlo code for Doppler effect calculations.
- (2) Comparative analysis of computation methods, successive- k , perturbation, and Monte Carlo.

- (3) Investigation of the energy and temperature heterogeneity effect on the Doppler coefficient with the improved Monte Carlo programme.
- (4) Effect of scattering resonances on the Doppler effect.
- (5) Resonance data and cross-sections for DE calculations (including high Pu isotopes).
- (6) The importance of multi-level analysis for Doppler calculations.
- (7) One-dimensional models for two-dimensional reactors for DE calculations.

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